

# Business Requirements Document (BRD)

## *Customer Lifetime Value & Churn Segmentation Analysis*

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|-------------|-------------------------------------------------------|
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| Project     | Customer Lifetime Value & Churn Segmentation Analysis |

### 1. Purpose

This document defines the business objectives, scope, and expected outcomes for a customer analytics project aimed at identifying high-value customers, measuring churn risk, and guiding retention decisions for an e-commerce business.

### 2. Business Problem

The business currently has no structured way to identify which customers contribute the most revenue or which valuable customers are at risk of disengaging. Retention efforts, where they happen, are not prioritized by customer value, resulting in inefficient use of retention resources.

### 3. Business Objectives

- Identify which customers drive the majority of revenue.
- Detect customers who were previously active and valuable but have stopped purchasing.
- Provide a prioritized, actionable list of at-risk customers for retention outreach.
- Distinguish between customers who spend through frequent small orders versus occasional large orders, to inform the right retention approach for each.

### 4. Scope

#### In Scope

- Segmentation of customers using Recency, Frequency, and Monetary (RFM) analysis.
- A churn classification model to estimate likelihood of customer disengagement.
- An interactive dashboard summarizing segments, revenue concentration, and an at-risk customer action list.

#### Out of Scope

- Execution of retention campaigns (this project informs targeting only, not outreach).
- Real-time/live data integration — analysis is based on a historical dataset.

### 5. Stakeholders

| Stakeholder                | Interest                                                                  |
|----------------------------|---------------------------------------------------------------------------|
| Retention / Marketing Team | Uses the at-risk customer list to plan outreach and campaigns.            |
| Business Leadership        | Uses revenue-concentration and churn findings to guide budget priorities. |
| Data/Analytics Team        | Maintains and extends the analysis and dashboard over time.               |

## 6. Success Criteria

- Customers are grouped into clear, distinct segments based on value and activity.
- The at-risk, high-value customer segment is clearly identified and quantified by revenue impact.
- A dashboard exists that a non-technical stakeholder can use without needing to understand the underlying analysis.

## 7. Assumptions & Constraints

- Analysis is based on a fixed historical dataset, not a live data feed.
- A 90-day inactivity window is used to define churn; this is a business assumption and can be adjusted.
- Only completed ("delivered") orders are included in customer value calculations.

## 8. Deliverables

- SQL data extraction and transformation scripts.
- Python scripts for RFM scoring, segmentation, and churn modelling.
- Interactive BI dashboard (Power BI).
- Summary document of findings and recommendations.